Covariant entropy accounting for interacting matter–plenum sectors on compact spatial manifolds

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Energy exchange between cosmological sectors is not, by itself, entropy production. We derive the distinction for interacting perfect-fluid sectors on a spacetime with compact spatial sections. If $\nabla_\mu T_A^{\mu\nu} = Q_A^\nu$, $\nabla_\mu(n_A u^\mu) = n_A\Gamma_A$, and $q_A \equiv -u_\nu Q_A^\nu$, the Gibbs relation gives

$$\nabla_\mu S_A^\mu = \frac{q_A - \mu_A n_A \Gamma_A}{T_A}.$$

Total stress conservation enforces $\sum_A q_A = 0$ in a common comoving frame, but does not fix the sign of $\sum_A \nabla_\mu S_A^\mu$. For two particle-conserving sectors the result is $q(1/T_1 - 1/T_2)$, which is nonnegative only after a constitutive direction for the transfer has been supplied. Compactness removes net spatial boundary flux; it does not create an external energy or entropy source.

We apply this accounting to the conditional R4C1 matter–plenum–reservoir action. Its action-derived currents and homogeneous balance equations support finite reversible exchange, but the model supplies no temperatures, chemical potentials, particle-production rates, dissipative transport matrix, or coarse-graining prescription. Consequently it does not yet derive positive entropy production, heat-death avoidance, a thermodynamic stability Hessian, or cosmic acceleration. We formulate the constitutive, stability, and observational conditions that a positive claim must satisfy. The result is a bounded thermodynamic framework and a set of no-go statements, not a completed syntropic cosmology.

I. PROBLEM AND SCOPE

Interacting-sector cosmologies commonly introduce currents through $\nabla_\mu T_A^{\mu\nu} = Q_A^\nu$. Such equations are necessary for consistent energy–momentum bookkeeping, but neither a nonzero Q_A^ν nor the identity $\sum_A Q_A^\nu = 0$ is a second-law statement. Relativistic non-equilibrium thermodynamics requires an entropy current and constitutive relations in addition to stress-energy balance [1–3].

This distinction is especially important for models that use the label *syntropic*. The name does not determine the direction of energy flow, the sign of entropy production, or the stability of an equilibrium. Those are calculations.

Our objective is methodological: to separate what follows from covariant balance identities from the additional physical input needed to assign entropy, prove a second-law sign, establish equilibrium stability, or infer cosmic acceleration. Sections II–IV are model independent within the stated local-equilibrium perfect-fluid assumptions. Section V then uses one conditional action as a diagnostic application; the general entropy identity does not depend on that model.

Table I records the resulting claim ladder.

II. COVARIANT SECTOR BALANCES

Consider sectors $A = 1, \ldots, N$ with symmetric stresses $T_A^{\mu\nu}$ and transfer currents

$$\nabla_\mu T_A^{\mu\nu} = Q_A^\nu, \qquad \sum_A Q_A^\nu = 0. \tag{1}$$

The second relation follows from total diffeomorphism invariance after all field equations are imposed; it should not be inserted as a substitute for varying an incompletely specified action.

Let $u^\mu u_\mu = -1$, $h^\mu{}_\nu = \delta^\mu{}_\nu + u^\mu u_\nu$, and decompose each current as

$$q_A \equiv -u_\nu Q_A^\nu, \qquad f_A^\mu \equiv h^\mu{}_\nu Q_A^\nu, \qquad Q_A^\mu = q_A u^\mu + f_A^\mu. \tag{2}$$

For a perfect fluid,

$$T_A^{\mu\nu} = (\rho_A + p_A)u^\mu u^\nu + p_A g^{\mu\nu}, \tag{3}$$

the timelike projection of Eq. (1) is

$$\dot\rho_A + \theta(\rho_A + p_A) = q_A, \qquad \theta \equiv \nabla_\mu u^\mu. \tag{4}$$

In an FLRW chart, $\theta = 3H$. Equation (4) fixes an energy transfer convention: $q_A > 0$ means energy enters sector A in the chosen comoving frame. The spatial projection gives its force density and must be retained in perturbative or non-comoving applications.

Summing Eq. (4) gives

$$\dot\rho_{\rm tot} + 3H(\rho_{\rm tot} + p_{\rm tot}) = 0, \tag{5}$$

for a common FLRW velocity. This identity is conservation, not an equation for entropy.

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Repository: `github.com/brendohxd/ITSM-...`

TABLE I. Status of thermodynamic claims in the present conditional model.

Claim	Status	Evidence or missing input
Sector exchange currents and total balance	Conditional derivation	R4C1 action variation and homogeneous on-shell control
Finite, reversible exchange on a sampled background	Locally checked	Both interface currents are nonzero and the reservoir transfer reverses sign
Positive total entropy production	Not derived	No T_A, μ_A, Γ_A, dissipative kernel, or coarse graining
Topology supplies external energy/entropy inflow	Excluded	$\int_{T^3} \sqrt{\gamma}\, D_i J^i = 0$ for smooth periodic J^i
Thermodynamic stability	Open	No constrained entropy Hessian or susceptibility matrix has been computed
Acceleration from exchange	Not derived	Requires the total pressure and a solved background; q_A cancels in the summed continuity equation
Avoidance of heat death	Not established	Requires a global state, asymptotics, and entropy definition beyond the present classical reversible action

III. GIBBS RELATION AND ENTROPY PRODUCTION

Introduce a particle current and specific entropy for each sector,

$$N_A^\mu = n_A u^\mu, \qquad \nabla_\mu N_A^\mu = n_A \Gamma_A, \qquad S_A^\mu = n_A \sigma_A u^\mu. \tag{6}$$

Here Γ_A is a fractional particle-production rate. It is independent information: an energy-transfer current does not determine it.

For local equilibrium the Gibbs relation per particle is

$$T_A \mathrm{d}\sigma_A = \mathrm{d}\left(\frac{\rho_A}{n_A}\right) + p_A\, \mathrm{d}\left(\frac{1}{n_A}\right). \tag{7}$$

Taking the derivative along u^μ and using Eqs. (4) and (6) yields

$$n_A T_A \dot{\sigma}_A = q_A - (\rho_A + p_A)\Gamma_A. \tag{8}$$

The Euler relation $\rho_A + p_A = T_A n_A \sigma_A + \mu_A n_A$ then gives the entropy-current divergence

$$\boxed{\nabla_\mu S_A^\mu = \frac{q_A - \mu_A n_A \Gamma_A}{T_A}} \tag{9}$$

for the idealized perfect-fluid sector. Heat flux, viscosity, diffusion, chemical reactions, and relaxation add further flux–force products; causal closures require more than the algebraic first-order form [1].

A. Two particle-conserving sectors

Set $\Gamma_1 = \Gamma_2 = 0$, $q_1 = q$, and $q_2 = -q$. Then

$$\nabla_\mu (S_1^\mu + S_2^\mu) = q\left(\frac{1}{T_1} - \frac{1}{T_2}\right). \tag{10}$$

The right-hand side can be positive, zero, or negative for an arbitrarily assigned q. A simple Onsager closure

$$q = L\left(\frac{1}{T_1} - \frac{1}{T_2}\right), \qquad L \geq 0, \tag{11}$$

gives

$$\nabla_\mu S_{\mathrm{tot}}^\mu = L\left(\frac{1}{T_1} - \frac{1}{T_2}\right)^2 \geq 0. \tag{12}$$

Thus nonnegative entropy production is a property of a constitutive law and its coefficients, not of $\sum_A Q_A^\nu = 0$.

B. Particle creation

When $\Gamma_A \neq 0$, chemical potentials and particle source terms enter Eq. (9). Constant specific entropy $\dot{\sigma}_A = 0$ imposes

$$q_A = (\rho_A + p_A)\Gamma_A, \tag{13}$$

but this is a constitutive restriction rather than a consequence of the continuity equation. Cosmological matter-creation thermodynamics has long made this distinction explicit [2, 4].

IV. COMPACT TOPOLOGY: WHAT VANISHES AND WHAT DOES NOT

Let each spatial slice be a smooth flat T^3 with periodic fields and induced metric γ_{ij}. For any smooth periodic spatial vector,

$$\int_{T^3} \mathrm{d}^3x\, \sqrt{\gamma}\, D_i J^i = 0. \tag{14}$$

This is the divergence theorem on a compact manifold without boundary. It forbids interpreting a pure divergence as net inflow from an exterior.

For a homogeneous entropy current, the integrated entropy balance is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3 V_0 \sum_A n_A \sigma_A\right) = a^3 V_0 \sum_A \frac{q_A - \mu_A n_A \Gamma_A}{T_A}, \quad (15)$$

where V_0 is the coordinate volume of the fundamental domain. Compactness removes a boundary term from this equation; it does not set the right-hand side to zero and does not choose its sign. Internal energy transfer and local entropy production remain possible.

The same boundary fact also makes zero modes and global constraints important. A spatially constant mode is not fixed by Eq. (14); its state, charge, and thermodynamic interpretation require separate input.

A. Provenance of the topology correction

Earlier ITSM working notes asserted that compact T^3 topology requires an inward energy stream and that matter–plenum exchange exactly cancels entropy growth. Both statements are withdrawn. Equation (14) excludes the proposed exterior boundary flux, while Eq. (9) shows why total energy conservation alone cannot cancel entropy production. The previous v0.6.0 source and PDF remain preserved as provenance; the present manuscript supersedes their thermodynamic claims.

V. APPLICATION TO THE CONDITIONAL R4C1 ACTION

A. Why R4C1 is used as a stress test

The general accounting boundary becomes operational when applied to a model whose interface currents are independently obtained from an action. R4C1 is useful here because it contains several dynamical sectors and nonzero, action-derived exchange currents, including a reservoir transfer that changes sign on the registered trajectory, while supplying no dissipative constitutive closure. It is therefore sufficiently specified to test conservation and finite reversible exchange, but deliberately insufficient to infer irreversible entropy production. This application diagnoses the gap between those statements; it is not evidence that R4C1 is a completed or observationally viable cosmology.

B. Action-derived currents

The R4C1 candidate is a closed, reversible, classical field action on $I \times T^3$. It contains a conformally coupled dust sector, a complex amplitude/phase sector with an independent force chart, a constrained timelike frame, and a real reservoir field. The baseline stress split gives the action-derived interface currents

$$Q^\nu_{\mathrm{mp}} = \beta T_m \nabla^\nu \psi, \quad (16)$$

$$Q^\nu_{\mathrm{syn}} = -\frac{g_r}{2} s\, r\, \nabla^\nu r, \qquad s = u^2 + v^2. \quad (17)$$

On all field equations the corresponding sector identities are

$$\nabla_\mu T^{\mu\nu}_m = Q^\nu_{\mathrm{mp}}, \quad (18)$$

$$\nabla_\mu T^{\mu\nu}_P = -Q^\nu_{\mathrm{mp}} + Q^\nu_{\mathrm{syn}}, \quad (19)$$

$$\nabla_\mu T^{\mu\nu}_R = -Q^\nu_{\mathrm{syn}}. \quad (20)$$

Their sum vanishes.

On the registered homogeneous background these reduce to

$$\dot\rho_m + 3H\rho_m = Q^0_{\mathrm{mp}}, \quad (21)$$

$$\dot\rho_P + 3H(\rho_P + p_P) = -Q^0_{\mathrm{mp}} + Q^0_{\mathrm{syn}}, \quad (22)$$

$$\dot\rho_R + 3H(\rho_R + p_R) = -Q^0_{\mathrm{syn}}. \quad (23)$$

The numerical control has nonzero interface currents and a finest independently differenced sector-balance residual of 7.96×10^{-8}. The reservoir transfer changes sign over the sampled trajectory. This is evidence for finite reversible energy exchange in the conditional model, not for an irreversible pump.

C. Why no entropy-production claim follows

The R4C1 action does not currently define:

1. sector temperatures T_m, T_P, T_R;
2. chemical potentials or particle currents for all sectors;
3. nonzero production rates Γ_A;
4. a dissipative response matrix such as L_{AB};
5. a coarse-graining map from the reversible field dynamics to macroscopic entropy; or
6. a quantum state and renormalized entropy for the compact modes.

Consequently Eq. (9) cannot yet be evaluated.

There is a second caution. Assigning a fraction η of the portal potential to a different sector changes the named reservoir current by $\eta\nabla^\nu W$ while leaving the total action and stress unchanged. A sector entropy source built from a split-dependent current is not an observable until the subsystem convention and coarse graining are physically fixed.

The action also conserves its full global $U(1)$ current. A nonzero condensate-number source has not been derived. Energy exchange and charge creation must not be conflated.

VI. THERMODYNAMIC STABILITY IS A CONSTRAINED HESSIAN PROBLEM

The second law and stability are different tests. A nonnegative instantaneous entropy-production rate does not prove that an equilibrium is a local maximum of entropy.

Let X^a denote independent extensive variables after all conserved totals and constraints have been imposed. Local thermodynamic stability requires

$$H^{(S)}_{ab} \equiv \frac{\partial^2 S_{\rm tot}}{\partial X^a \partial X^b} \preceq 0 \tag{24}$$

on the allowed tangent space. Equivalently, an energy representation requires the appropriate susceptibility matrix to be positive semidefinite.

For two subsystems exchanging only energy at fixed total energy,

$$\frac{\mathrm{d}^2 S_{\rm tot}}{\mathrm{d}E_1^2} = -\frac{1}{T_1^2 C_1} - \frac{1}{T_2^2 C_2}. \tag{25}$$

Positive heat capacities are sufficient for stability along this one transfer direction, but a coupled matter–plenum system can have additional density, charge, amplitude, phase, and modulus directions. Each physical eigenvalue of the constrained Hessian must be reported over a registered domain.

The current R4C1 quadratic-action work concerns dynamical kinetic, gradient, constraint, and mass matrices. Those are necessary for field stability but are not interchangeable with Eq. (24). No numerical thermodynamic Hessian eigenvalues are claimed here.

VII. EXCHANGE DOES NOT BY ITSELF CAUSE ACCELERATION

In Einstein gravity an FLRW background obeys

$$\frac{\ddot{a}}{a} = -\frac{1}{6M_P^2}\sum_A (\rho_A + 3p_A). \tag{26}$$

The transfer terms cancel from the summed continuity equation (5). Exchange may alter the sector partition and hence the pressures, but acceleration requires a solved stress history satisfying Eq. (26). Entropy production, an acceleration transition, and thermodynamic stability are therefore three distinct questions.

This separation has become observationally concrete. A recent background analysis of thermogravity found that acceleration generated solely by universal energy non-conservation is strongly disfavored by supernovae plus DESI BAO, with a reported $\Delta\chi^2 \simeq +51$ relative to flat ΛCDM [5]. R4C1 is not that model: its total stress is conserved. The comparison nevertheless shows why an exchange mechanism cannot be promoted from formal consistency to cosmological success without deriving $H(z)$, the ruler calibration, and the observational dictionary.

VIII. METHODOLOGICAL CONDITIONS FOR SYNTROPIC COSMOLOGICAL CLAIMS

To avoid conflating conservation identities with thermodynamic closure, a model in the interacting-sector class considered here that advances a positive syntropic claim must provide the following model-specific ingredients and tests:

1. a declared physical subsystem split and equations of state $\rho_A(n_A, \sigma_A, \ldots)$;
2. temperatures, chemical potentials, entropy currents, and particle currents derived from that state space;
3. action-derived or microphysically derived Γ_A where particle creation is claimed;
4. a causal constitutive closure for heat, diffusion, viscosity, chemical reactions, and reservoir transfer, with a positive entropy-production form;
5. the constrained entropy Hessian and all eigenvalues over a registered domain;
6. compatibility between thermodynamic and quadratic-action stability, including constraint singularities.

These conditions are necessary within the local-equilibrium framework used here; they are not jointly sufficient, and alternative thermodynamic formalisms must supply equivalent physical closure. If a claim extends to cosmic acceleration or observational viability, it additionally requires the full action-derived $H(z)$, $q(z)$, sound horizon, BBN history, and perturbations, followed by common-prior likelihood comparisons. Independent external review remains a publication requirement, not a substitute for any physical item above.

A failure of any positivity, causality, uniqueness, or observational test must remain visible. Conversely, passing a conservation residual or a Hessian check alone would not establish the rest of the chain.

IX. CONCLUSIONS

The covariant entropy identity (9) supplies the missing accounting boundary for interacting ITSM sectors. It establishes four points.

First, total stress conservation and entropy production are logically independent. Second, compact T^3 topology removes external boundary flux rather than supplying it. Third, R4C1 presently realizes finite, action-derived, reversible exchange but does not contain the constitutive data needed to prove positive entropy production or particle creation. Fourth, thermodynamic stability, dynamical field stability, and cosmic acceleration are separate calculations.

The resulting audit sequence is portable across interacting-sector models: derive the currents, specify the thermodynamic state space and constitutive closure, evaluate entropy production, test the constrained thermodynamic and field-stability spectra, and only then confront cosmological observables. R4C1 currently reaches the first step and a bounded reversible-exchange control.

Within that scope, this work provides a covariant framework in which proposals for entropy reduction, heat-death avoidance, or exchange-driven acceleration can be made falsifiable without confusing conservation identities with thermodynamic closure. It establishes none of those positive mechanisms in the present conditional model.

REPRODUCIBILITY AND PROVENANCE

The conditional action is frozen in `RES001_R4C1_ACTION_FREEZE_2026-09-25.md` (SHA-256 prefix `81bfc2e4f17b`). The first-variation report containing Eq. (17) has prefix `4834ea7517fd`; the interacting background report containing Eq. (23) has prefix `48b49fddd906`. Full digests, contracts, executables, and JSON receipts are retained in the companion repository. Hashes establish artifact identity, not physical truth or independent review.

DECLARATION ON THE USE OF ARTIFICIAL-INTELLIGENCE TOOLS

Artificial-intelligence-assisted tools were used for editorial drafting, source-code assistance, and checks of symbolic and numerical consistency. The author independently reviewed the manuscript, equations, cited sources, and generated artifacts and accepts full responsibility for the content.

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